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Input	Output
A	C
1	0
0	1

Table 2.9 Truth Table for NOT Gate

2.9.4 NOR Gate

The NOR gate circuit is an OR gate followed by an an inverter. Its output is "true" if both inputs are "false" Otherwise, the output is "false". In other words, the only way to get '1' as output is to have both inputs '0'. Otherwise the output is 0. The logic circuit of the NOR gateis

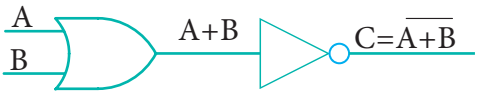

Fig. 2.7 Logic Circuit of NOR Gate

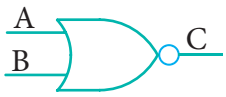

Fig. 2.8 Logic symbol of NOR Gate

The output of NOR gate is

$$C = (\overline{A + B})$$

Read this as "C equals NOT of A OR B" or "C equals the complement of A OR B".

For example if both the inputs are 0,

$$C = (\overline{0 + 0}) = \overline{0} = 1$$

The truth table for NOR gate is

Input		Output
A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

Table 2.10 Truth Table for NOR Gate

2.9.5 Bubbled AND Gate

The Logic Circuit of Bubbled AND Gate

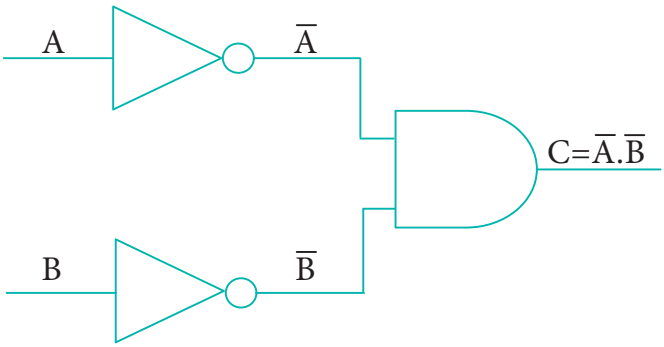

Fig. 2.9 Logic circuit of Bubbled AND Gate

In the above circuit, invertors on the input lines of the AND gate gives the output as

$$C = (\overline{A} \cdot \overline{B})$$

This circuit can be redrawn as the bubbles on the inputs, where the bubbles represent inversion.

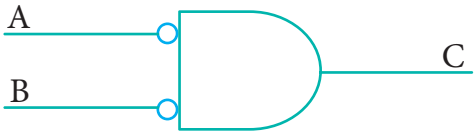

Fig. 2.10 Logic Symbol of Bubbled AND Gate

We refer this as bubbled AND gate. Let us analyse this logic circuit for all input possiblities.